Open problems:
Schrödinger equation is equivalent to information conservation

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`).

From the brief

1. **Classical counterpart.** State and prove the classical analogue: Hamiltonian evolution is equivalent to preservation of the Shannon entropy and of "the definitional independence of degrees of freedom". (Brief, Sec. 4: "This parallels the classical case where Hamiltonian evolution is equivalent to the preservation of Shannon entropy (Liouville's theorem)".)

2. **Entropy conservation as determinism and reversibility.** The brief reads DR-INFO as "an information theoretic way to characterize determinism and reversibility". A precise definition of deterministic and reversible evolution, and a proof that it is equivalent to DR-INFO, are not given. (Brief, Sec. 1.)

3. **Propagator for time-dependent $H(t)$.** "The Schrödinger equation gives rise to a differentiable unitary propagator $U_{t_0 \to t_1}$" is assumed. The formalization proves existence for constant H only; for bounded $H(t)$ it is the linear ODE existence theorem, and for unbounded self-adjoint $H(t)$ it needs Kato-type hypotheses. The converse has the same gap: for unbounded $H(t)$, Step 3 gives only a symmetric $H(t_0)$ (next section, first item). (Brief, Sec. 3, proof of Theorem 2 ($\Rightarrow$); Log, "Differences from the brief".)

From the formalization

Beyond finite dimension the formalization has a part in any dimension; it leaves open the items "Unbounded time-dependent Hamiltonians", "Norm-differentiable $U(t)$ in infinite dimension" and "Wold–Cooper reduction" below.

1. **Unbounded time-dependent Hamiltonians.** EVDIFF on every density operator forces a bounded H (Banach–Steinhaus). With EVDIFF on a dense set of states, Theorem 2 is formalized for a time-independent evolution, with unbounded self-adjoint H (`drSceq_iff_drInfo_and_evDiff`, by Stone's theorem). Without EVTI, Step 3 gives only a skew-symmetric $A(t_0)$, and $H(t_0)$ need not be self-adjoint: the propagator of $H(t) = H_\beta$ for $t < 0$ and H_α for $t \geq 0$, two self-adjoint extensions of the momentum on $L^2(0,1)$, is differentiable at 0 exactly on the domain of the minimal momentum operator, which is not self-adjoint. Open: the hypotheses (Kato-type, or on the evolution) under which DR-INFO + EVDIFF gives a self-adjoint $H(t)$, and the formalization of the example. (Log, "Unbounded Hamiltonians".)

2. **Equation at every time.** The formalization gives $U(t)$ only for t near t_0 and the Schrödinger equation only at t_0. A global differentiable unitary propagator $U_{t_0 \to t_1}$ for all times, with phases chosen consistently with $\Phi_{t_1 \to t_2} \circ \Phi_{t_0 \to t_1} = \Phi_{t_0 \to t_2}$, is not built. (Log, "Differences from the brief", item on `schrodinger_of_preserves`.)

3. **Norm-differentiable $U(t)$ in infinite dimension.** In any dimension only the strong equation $\psi'(t_0) = (i\hbar)^{-1} H\psi(t_0)$ is proved; the phase-fixing basis sum does not converge in operator norm. Whether a phase choice makes $t \mapsto U(t)$ differentiable in operator norm is open. (Log, any-dimension part, "Differences from the brief", EVDIFF item.)

4. **Wold–Cooper reduction.** The right shifts V_t on $L^2(0, \infty)$ give an evolution that satisfies [B]EVMIX, maps pure states to pure states, preserves transition probabilities and the entropy, but is induced by no unitary; this is formalized (`shift_semigroup_counterexample`, in the brief "Multiple equivalent characterizations of the Schrödinger equation"). The Wold–Cooper decomposition, which reduces every strongly continuous semigroup of isometries to a unitary group plus copies of this one, is not formalized. (Log, any-dimension part, "Problems found in the brief", second item.)